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Task Overview

The objective of this task was to build a simple rule-based conversational chatbot in Python. The chatbot interacts continuously with the user, recognizes different messages, provides appropriate responses, and allows the user to exit the conversation gracefully.

Approach

- The chatbot first asks for the user's name before starting the main conversation.
- A while True loop keeps the conversation running continuously.
- input() is used to receive messages from the user, and .lower().strip() cleans the input so that responses do not depend on capitalization or extra spaces.
- The in operator is used (for example, checking if "hello" or "hi" is present in the message) so that variations like "hello bot" or "hey hello" still trigger the correct response.
- if / elif / else statements handle the decision-making for each type of message.
- The loop is exited using break when the user types "bye" or "exit".
- The user's name is used in several responses to make the conversation feel personal.
- An else block provides a helpful fallback response for any unrecognized message.

Python Source Code

```
# Simple Rule-Based Python Chatbot
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def main():
    print("Chatbot: Hello! Before we start, what is your name?")
    name = input("You: ").lower().strip()
    name = name.title()

    print(f"Chatbot: Nice to meet you, {name}! Let's chat. Type 'bye' or 'exit' anytime to quit.")

    while True:
        message = input("You: ").lower().strip()

        if "bye" in message or "exit" in message:
            print(f"Chatbot: Goodbye, {name}! Have a great day!")
            break

        elif "hello" in message or "hi" in message:
            print(f"Chatbot: Hello {name}! How can I help you today?")

        elif "how are you" in message:
            print("Chatbot: I'm just a bunch of code, but I'm doing great! How about you?")

        elif "your name" in message:
            print("Chatbot: I'm ChatBot, your friendly Python assistant.")

        elif "time" in message:
            print("Chatbot: Sorry, I can't check the real time yet, but it's always a good time to code!")

        elif "help" in message:
```

```

        print(f"Chatbot: Sure {name}, you can ask me about my name, how I am, or just say hello!")

    elif "thank" in message:
        print("Chatbot: You're welcome!")

    else:
        print("Chatbot: Sorry, I didn't understand that. Could you rephrase?")

if __name__ == "__main__":
    main()

```

Sample Test Run

The chatbot was tested with the inputs suggested in the task sheet, including repeated and capitalized greetings, a recognized question, an unknown message, and the exit command:

```

Chatbot: Hello! Before we start, what is your name?
You: Rukesh
Chatbot: Nice to meet you, Rukesh! Let's chat. Type 'bye' or 'exit' anytime to quit.
You: hello
Chatbot: Hello Rukesh! How can I help you today?
You: HELLO
Chatbot: Hello Rukesh! How can I help you today?
You: hi
Chatbot: Hello Rukesh! How can I help you today?
You: what is your name
Chatbot: I'm ChatBot, your friendly Python assistant.
You: asdkjaskd
Chatbot: Sorry, I didn't understand that. Could you rephrase?
You: bye
Chatbot: Goodbye, Rukesh! Have a great day!

```

Requirements Checklist

- Asks the user's name before the main conversation – Done
- Uses while True: to keep the conversation running – Done
- Uses input() to receive messages – Done
- Cleans input using .lower().strip() – Done
- Recognizes at least two greetings (hello, hi) – Done
- Answers at least three different questions/topics (name, how are you, time, help, thanks) – Done
- Uses if / elif / else for decision making – Done
- Uses break to exit on bye / exit – Done
- Uses the user's name in responses – Done
- Provides a useful fallback response using else – Done
- Challenge: uses the in operator for flexible matching – Done

Conclusion

This task helped in understanding how loops, conditionals, and string methods can be combined to build a basic interactive program. The chatbot follows the INPUT → PROCESS → RESPOND → REPEAT → EXIT flow described in the task sheet and satisfies all the listed requirements.